

Supplemental Material for “Prescribed finite-window reaction-time modality by conserved-budget support design”

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This Supplemental Material states the bounded and natural-decay realisations of the Doi generator, proves compact-positive-time convergence of every prescribed finite time/control mixed derivative of the budget-normalized reaction density to the free-exposure law, and supplies the complete proof of exact prescribed finite modality. The construction is pointwise in a fixed finite physical dimension and a fixed finite mode count. It fixes a sufficiently small noise-and-width scale before choosing a sufficiently small positive reaction budget. The result is an existence theorem for a mode-dependent support design and does not assert switching among mode counts on one fixed support family.

S1. DOI MODEL AND ANALYTIC REALIZATIONS

For a fixed finite physical dimension $d \geq 2$, the longitudinal-slab symmetry reduction gives

$$\mathcal{Q}_d^{\text{box}} = I_z \times I_{\parallel} \times \mathbb{T}_W^{d-1}, \quad \mathcal{Q}_d^{\infty} = \mathbb{R} \times \mathbb{R} \times \mathbb{T}_W^{d-1}. \quad (\text{S1})$$

With $x = (z, r_{\parallel}, r_{\perp})$, equal particle diffusivities give the free forward operator

$$\mathcal{L}q = -\nabla \cdot (bq) + \nabla \cdot (\mathbf{D} \nabla q), \quad \mathbf{D} = \text{diag}(D/2, 2D, \dots, 2D), \quad (\text{S2})$$

where

$$b(x) = (-\gamma(z - \bar{z}), -\gamma r_{\parallel}, 0, \dots, 0). \quad (\text{S3})$$

The bounded quotient has zero forward flux on its nonperiodic faces and periodic transverse coordinates. Natural decay on the unbounded quotient is specified by the reversible density

$$\pi(x) = Z_{\pi}^{-1} \exp \left[-\frac{\gamma(z - \bar{z})^2}{D} - \frac{\gamma r_{\parallel}^2}{4D} \right], \quad (\text{S4})$$

including the normalized uniform torus factor.

Let $0 < a < W/2$, let $\chi_a(r)$ be the embedded minimum-image contact-ball indicator, and let normalized nonnegative longitudinal profiles obey $\phi_j \in L^{\infty}$ and $\int \phi_j(z) dz = 1$. Put

$$V_j(z, r) = W^{-(d-1)} \chi_a(r) \phi_j(z), \quad V_w = \sum_{j=1}^J w_j V_j, \quad K_{B,w} = B V_w, \quad (\text{S5})$$

where $w_j \geq 0$ and $\sum_j w_j = 1$. The installed centre-space catalyst amount is B for every w . The killed state and normalized reaction density are

$$q_{B,w}(t) = e^{t(\mathcal{L} - B M_{V_w})} q_0, \quad F_B(t, w) = \langle V_w, q_{B,w}(t) \rangle. \quad (\text{S6})$$

Proposition S1.1 (Analytic killed semigroups). *On $\mathcal{Q}_d^{\text{box}}$, the reflecting/periodic realization of $\mathcal{L}$ on $L^2(\mathcal{Q}_d^{\text{box}})$ generates a positive analytic semigroup. On $\mathcal{Q}_d^{\infty}$, the natural-decay realization on $X_{\pi} = L^2(\pi^{-1} dx)$ does the same. For real $B \geq 0$ and simplex control w , bounded multiplication by $-B V_w$ preserves the generator domain and yields a positive, mass-decreasing analytic semigroup. At positive time the state is entire in every finite affine complexification of the control amplitudes. On $\mathcal{Q}_d^{\infty}$, the observable pairing is continuous when $V_w \in L^2(\pi dx) \cap L^{\infty}$.*

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Proof. Set $q = \pi u$. The identity

$$\mathcal{L}(\pi u) = \pi \mathcal{G} u, \quad \mathcal{G} = \pi^{-1} \nabla \cdot (\pi \mathbf{D} \nabla) \quad (\text{S7})$$

follows from $b = \mathbf{D} \nabla \log \pi$. On $L^2(\pi dx)$, the closed weighted form has

$$\langle u, \mathcal{G} v \rangle_{L^2(\pi)} = - \int (\nabla u)^\top \mathbf{D} \nabla v \pi dx. \quad (\text{S8})$$

It therefore realizes $\mathcal{G}$ as a nonpositive self-adjoint generator. Multiplication by π is a bounded isomorphism on the bounded quotient and a unitary map into X_π on the unbounded quotient. Bounded multiplication by $-BV_w$ preserves analyticity and the domain. The norm-convergent Dyson–Phillips series gives entire dependence on the finite control amplitudes. Positivity and mass decrease for real nonnegative data follow from the product formula, equivalently from Feynman–Kac [1, 2]. $\square$

S2. UNIFORM COMPACT-POSITIVE-TIME MIXED DERIVATIVES

Choose an interior affine control chart $w(\theta) = w_* + P\theta$ and a compact real control set Θ . Fix $\delta > 0$ such that each closed coordinate polydisc of radius δ about a point of Θ lies in the complex tube

$$\Theta_\delta^+ = \{\zeta \in \mathbb{C}^{J-1} : \text{dist}_\infty(\zeta, \Theta) < 2\delta\}. \quad (\text{S9})$$

Define

$$G(t, \theta) = \langle V_{w(\theta)}, e^{t\mathcal{L}} q_0 \rangle, \quad v_{\infty, \delta} = \sup_{\zeta \in \Theta_\delta^+} \|V_{w(\zeta)}\|_\infty. \quad (\text{S10})$$

Let $v_{*, \delta}$ be the supremum of $\|V_{w(\zeta)}\|_{L^2}$ on the bounded quotient and of $\|V_{w(\zeta)}\|_{L^2(\pi dx)}$ on the unbounded quotient. Write κ_π for the bounded-quotient similarity bound and set $\kappa_\pi = 1$ on the unbounded quotient.

Theorem S2.1 (Uniform weak-reaction mixed-jet bridge). *Fix $0 < \tau \leq T < \infty$, $0 \leq B \leq B_{\max}$, an integer $r \geq 0$, and a finite control multi-index α . The normalized density $F_B(t, \theta)$ in Eq. (S6) extends continuously to $F_0 = G$, and*

$$\begin{aligned} & \sup_{(t, \theta) \in [\tau, T] \times \Theta} |\partial_t^r \partial_\theta^\alpha (F_B - G)| \\ & \leq r! \alpha! \left(\frac{2}{\tau}\right)^r \delta^{-|\alpha|} v_{*, \delta} \kappa_\pi \left[\exp\left(\frac{3}{2} B v_{\infty, \delta} T\right) - 1 \right] \|q_0\|_X. \end{aligned} \quad (\text{S11})$$

In particular, the left side is at most $BC_{r, \alpha}$ for a finite constant independent of $B \in [0, B_{\max}]$.

Proof. For complex time z with $\text{Re } z > 0$, the norm-convergent Dyson expansion in the self-adjoint representation is

$$e^{z(\mathcal{G} - BV_w)} = e^{z\mathcal{G}} + \sum_{n=1}^{\infty} (-Bz)^n \int_{\Delta_n} e^{z(1-s_1)\mathcal{G}} V_w \left[\prod_{k=1}^{n-1} e^{z(s_k - s_{k+1})\mathcal{G}} V_w \right] e^{zs_n\mathcal{G}} d\mathbf{s}, \quad (\text{S12})$$

where $\Delta_n = \{1 \geq s_1 \geq \dots \geq s_n \geq 0\}$ has volume $1/n!$. Every free factor is contractive. Consequently,

$$\|e^{z(\mathcal{L} - BV_w)} - e^{z\mathcal{L}}\|_{X \rightarrow X} \leq \kappa_\pi \{e^{B|z|\|V_w\|_\infty} - 1\}. \quad (\text{S13})$$

The complex-bilinear observable remainder thus satisfies

$$\left| \left\langle V_{w(\zeta)}, [e^{z(\mathcal{L} - BV_{w(\zeta)})} - e^{z\mathcal{L}}] q_0 \right\rangle \right| \leq v_{*, \delta} \kappa_\pi \{e^{Bv_{\infty, \delta}|z|} - 1\} \|q_0\|_X. \quad (\text{S14})$$

For $t \in [\tau, T]$, the disk $|z - t| \leq \tau/2$ remains in the open right half-plane and has $|z| \leq 3T/2$. Cauchy's estimate on this disk and on each radius- δ control polydisc proves Eq. (S11). Finally, $e^{Bx} - 1 \leq Bxe^{B_{\max}x}$ gives the linear bound. $\square$

The estimate transfers any fixed finite collection of strict stationary-point neighbourhood, curvature, endpoint, and complement inequalities. It is asserted only on compact positive-time intervals and does not supply a useful numerical budget threshold.

S3. EXACT PRESCRIBED FINITE MODALITY: COMPLETE PROOF

This section proves the exact finite-mode statement used in the theorem-first working paper. The construction is pointwise in a fixed finite dimension and a fixed finite mode count. Its two small parameters are ordered: first the narrow-noise parameter is fixed sufficiently small, and only then is the Doi budget taken sufficiently small. All stationary-point counts below are on the declared compact positive-time window.

S3.1. Exact quotient, stationary midpoint variance, and whole-window contact

Fix finite integers $d \geq 2$ and $m \geq 1$, a transverse period $W > 0$, a contact radius $0 < a < W/2$, and a compact positive-time interval $I = [\tau, T] \subset (0, \infty)$. On $\mathbb{R} \times \mathbb{R} \times \mathbb{T}_W^{d-1}$, let the longitudinal midpoint and relative coordinates solve

$$dZ_t = -\gamma(Z_t - \bar{z}) dt + \varepsilon \sqrt{D_0} dW_t, \quad (\text{S15})$$

$$dR_{\parallel,t} = -\gamma R_{\parallel,t} dt + 2\varepsilon \sqrt{D_0} dW_{\parallel,t}, \quad (\text{S16})$$

$$dR_{\perp,t} = 2\varepsilon \sqrt{D_0} dW_{\perp,t} \pmod{W}, \quad (\text{S17})$$

where $D_0, \gamma > 0$. All Brownian drivers and all initial variables are mutually independent, and

$$Z_0 \sim N\left(z_0, \varepsilon^2 \frac{D_0}{2\gamma}\right), \quad z_0 \neq \bar{z}, \quad (\text{S18})$$

$$R_{\parallel,0} \sim N(r_{\parallel,0}, \varepsilon^2 u_0^2), \quad 0 < u_0^2 < \frac{4D_0}{\gamma}, \quad (\text{S19})$$

$$R_{\perp,0} \sim \text{WN}(r_{\perp,0}, \varepsilon^2 \Sigma_{\perp,0}), \quad \Sigma_{\perp,0} \succ 0. \quad (\text{S20})$$

Here WN denotes the wrapped Gaussian on the transverse torus. The two strict variance inequalities $D_0/(2\gamma) < D_0/\gamma$ and $u_0^2 < 4D_0/\gamma$ are precisely the unbounded Gaussian conditions that place this initial law in the weighted state space used by Theorem S2.1; the wrapped factor is square integrable against the uniform torus measure.

The midpoint has mean and variance

$$\mu(t) = \bar{z} + (z_0 - \bar{z})e^{-\gamma t}, \quad \text{Var } Z_t = \varepsilon^2 \frac{D_0}{2\gamma} \quad (t \geq 0). \quad (\text{S21})$$

Thus only the variance, not the mean, is stationary. This constant variance is what produces a common-variance Gaussian mixture below.

Let $\varsigma = \text{sgn } \mu' \in \{-1, 1\}$, fix a physical reference length $\ell_0 > 0$, and define

$$x(t) = \frac{\varsigma \mu(t)}{\ell_0}, \quad v_0 := \inf_{t \in I} x'(t) > 0. \quad (\text{S22})$$

Choose target times and their dimensionless centres by

$$\tau < t_1 < \cdots < t_m < T, \quad c_j = x(t_j). \quad (\text{S23})$$

Then $c_1 < \cdots < c_m$, and the physical centre of slab j is $\varsigma \ell_0 c_j = \mu(t_j)$. In particular, reversing the trajectory orientation reverses the physical centres at the same time; no centre is held fixed under the coordinate reversal.

Write

$$r_*(t) = (r_{\parallel,0} e^{-\gamma t}, r_{\perp,0}) \quad (\text{S24})$$

and impose the whole-window, minimum-image contact margin

$$\sup_{t \in I} |r_*(t)|_{\text{mi}} \leq a - \eta \quad \text{for some } \eta > 0. \quad (\text{S25})$$

The word “whole-window” is essential: the condition is required on every point of I , not merely near the target times.

Lemma S3.1 (Whole-window differentiated contact tail). *Let*

$$c_{d,\varepsilon}(t) = \mathbb{P}\{|R_t|_{\text{mi}} < a\}. \quad (\text{S26})$$

Under Eqs. (S16)–(S25), for $r = 0, 1, 2$ there are finite constants $C_{r,d}, N_{r,d}, q_d > 0$, independent of all sufficiently small ε , such that

$$\|\partial_t^r(c_{d,\varepsilon} - 1)\|_{L^\infty(I)} \leq C_{r,d}\varepsilon^{-N_{r,d}} \exp(-q_d/\varepsilon^2). \quad (\text{S27})$$

Consequently, $c_{d,\varepsilon} \geq 1/2$ for sufficiently small ε , and the first two logarithmic derivatives of $c_{d,\varepsilon}$ are uniformly bounded on I and converge to zero faster than any power of ε .

Proof. In an unwrapped transverse chart the relative Gaussian has mean $r_*(t)$ and covariance $\varepsilon^2 \Sigma_R(t)$, with longitudinal and transverse blocks

$$u_0^2 e^{-2\gamma t} + \frac{2D_0}{\gamma}(1 - e^{-2\gamma t}), \quad \Sigma_{\perp,0} + 4D_0 t I_{d-1}. \quad (\text{S28})$$

Because $I \subset (0, \infty)$, $u_0 > 0$, and $\Sigma_{\perp,0} \succ 0$, this covariance and its first two time derivatives are uniformly bounded on I , and its eigenvalues lie between two positive constants. The reverse triangle inequality and Eq. (S25) give

$$\inf_{t \in I} \inf_{|r|_{\text{mi}} \geq a} d_{\text{cyl}}(r, r_*(t)) \geq \eta. \quad (\text{S29})$$

The assumption $a < W/2$ keeps the contact ball away from the transverse cut locus. Express the wrapped density as its Gaussian lattice-image sum. Every time derivative of order at most two is the same Gaussian multiplied by a polynomial in the scaled displacement and in ε^{-1} , with degree bounded independently of t . Equation (S29), the uniform covariance bounds, and a standard Gaussian tail estimate therefore give, after summing the lattice images,

$$\sup_{t \in I} \int_{|r|_{\text{mi}} \geq a} |\partial_t^r p_{\varepsilon,t}(r)| dr \leq C_{r,d} \varepsilon^{-N_{r,d}} e^{-q_d/\varepsilon^2}, \quad r = 0, 1, 2. \quad (\text{S30})$$

Differentiation under the integral is justified by the same integrable majorant and yields Eq. (S27). For small ε , its $r = 0$ case gives $c_{d,\varepsilon} \geq 1/2$, while

$$\begin{aligned} |\partial_t \log c_{d,\varepsilon}| &\leq 2|\partial_t c_{d,\varepsilon}|, \\ |\partial_t^2 \log c_{d,\varepsilon}| &\leq 2|\partial_t^2 c_{d,\varepsilon}| + 4|\partial_t c_{d,\varepsilon}|^2. \end{aligned} \quad (\text{S31})$$

This proves the logarithmic-derivative statement. $\square$

Fix $\rho > 0$, put

$$S_*^2 = \frac{D_0}{2\gamma} + \rho^2, \quad \sigma = \frac{\varepsilon S_*}{\ell_0}, \quad (\text{S32})$$

and use the normalized physical slabs

$$\phi_{j,\varepsilon}(z) = \frac{1}{\sqrt{2\pi} \varepsilon \rho} \exp\left[-\frac{(z - \mu(t_j))^2}{2\varepsilon^2 \rho^2}\right]. \quad (\text{S33})$$

Let $0 < w_* \leq 1/m$ and

$$\mathcal{W}_{w_*} = \left\{ w = (w_1, \dots, w_m) : \sum_{j=1}^m w_j = 1, \quad w_j \geq w_* > 0 \right\}. \quad (\text{S34})$$

For $w \in \mathcal{W}_{w_*}$, define the Doi killing field

$$K_{B,w,\varepsilon}(Z, R) = \mathbf{1}_{\{|R|_{\text{mi}} < a\}} \frac{B}{W^{d-1}} \sum_{j=1}^m w_j \phi_{j,\varepsilon}(Z) = B V_{w,\varepsilon}(Z, R). \quad (\text{S35})$$

Every slab has unit full longitudinal integral, so B is the same installed centre-space budget for every allocation.

Lemma S3.2 (Exact common-variance free-exposure factorization). *Let $T_0(t)$ be the unkilld semigroup and q_0 the initial law above. The unit-budget free-exposure clock is exactly*

$$G_{\varepsilon,w}(t) := \langle V_{w,\varepsilon}, T_0(t)q_0 \rangle = a_\varepsilon(t)H_{\sigma,w}(x(t)), \quad (\text{S36})$$

$$a_\varepsilon(t) := \frac{c_{d,\varepsilon}(t)}{W^{d-1}\sqrt{2\pi}\varepsilon S_*}, \quad (\text{S37})$$

$$H_{\sigma,w}(x) := \sum_{j=1}^m w_j \exp\left[-\frac{(x - c_j)^2}{2\sigma^2}\right]. \quad (\text{S38})$$

The positive factor a_ε has uniformly bounded first two logarithmic time derivatives for all sufficiently small ε .

Proof. Independence factors the contact event from every longitudinal slab expectation. By Eq. (S21), convolution of $\phi_{j,\varepsilon}$ with the law of Z_t gives

$$\mathbb{E}[\phi_{j,\varepsilon}(Z_t)] = \frac{1}{\sqrt{2\pi}\varepsilon S_*} \exp\left[-\frac{(\mu(t) - \mu(t_j))^2}{2\varepsilon^2 S_*^2}\right]. \quad (\text{S39})$$

Since $\mu(t) - \mu(t_j) = \varsigma \ell_0(x(t) - c_j)$, Eqs. (S32) and (S39) give Eqs. (S36)–(S38). The last assertion follows from Lemma S3.1, because the remaining factors in a_ε are independent of time. $\square$

S3.2. The pure common-variance mixture

Set $J = x(I) = [\alpha, \beta]$. The targets are interior, so

$$\alpha < c_1 < \cdots < c_m < \beta. \quad (\text{S40})$$

For $m \geq 2$, let

$$\Delta_j = c_{j+1} - c_j, \quad \Delta_* := \min_{1 \leq j < m} \Delta_j > 0. \quad (\text{S41})$$

No empty minimum is used when $m = 1$.

Define

$$q_j(x) = w_j e^{-(x-c_j)^2/(2\sigma^2)}, \quad \pi_j(x) = \frac{q_j(x)}{H_{\sigma,w}(x)}, \quad \bar{c}(x) = \sum_{j=1}^m \pi_j(x) c_j. \quad (\text{S42})$$

The logarithmic slope $L_{\sigma,w} = \partial_x \log H_{\sigma,w}$ obeys the exact identities

$$L_{\sigma,w}(x) = \frac{\bar{c}(x) - x}{\sigma^2}, \quad (\text{S43})$$

$$L'_{\sigma,w}(x) = \frac{\text{Var}_{\pi(x)}(c)}{\sigma^4} - \frac{1}{\sigma^2}. \quad (\text{S44})$$

Indeed, differentiation of q_j gives the first identity, while $\pi'_j = \pi_j(c_j - \bar{c})/\sigma^2$ gives the second.

The mode cap for univariate Gaussian mixtures is classical [3, 4]. We retain the direct multiplicity-counting proof because the later Doi transfer also needs explicit uniform root locations, curvature scales, and full-complement margins.

Lemma S3.3 (Global extended-Chebyshev zero bound). *For distinct real centres and positive weights, $H'_{\sigma,w}$ has at most $2m - 1$ real zeros counted with multiplicity. Its real zero set is finite.*

Proof. Multiplication by a nowhere-zero function preserves both zeros and their multiplicities, and direct differentiation gives

$$\sigma^2 e^{x^2/(2\sigma^2)} H'_{\sigma,w}(x) = \sum_{j=1}^m w_j (c_j - x) e^{-c_j^2/(2\sigma^2)} e^{c_j x/\sigma^2}. \quad (\text{S45})$$

We prove the slightly more general claim that a nonzero function

$$P_m(x) = \sum_{j=1}^m (a_j + b_j x) e^{\lambda_j x}, \quad \lambda_1 < \cdots < \lambda_m, \quad (\text{S46})$$

with no identically zero summand has at most $2m - 1$ real zeros counted with multiplicity.

First, the zero set is finite. After multiplication by $e^{-\lambda_1 x}$, the first affine term dominates as $x \rightarrow -\infty$; the last affine-exponential term dominates as $x \rightarrow +\infty$. Each dominating affine factor has a fixed sign sufficiently far in its tail. Thus all zeros lie in a compact interval. A nonzero real-analytic function has only finitely many zeros there.

Now induct on m . The $m = 1$ case is the affine zero bound. For $m > 1$, put $Q = e^{-\lambda_1 x} P_m$. Two derivatives annihilate the first affine term. For every $j \geq 2$, the second derivative of $(a_j + b_j x)e^{(\lambda_j - \lambda_1)x}$ is a nonzero affine polynomial times the same exponential, and the remaining positive exponents are distinct. If a C^2 function has $N \geq 2$ real zeros counted with multiplicity, generalized Rolle counting gives at least $N - 2$ zeros of its second derivative counted with multiplicity: differentiation reduces the multiplicity at each root, and ordinary Rolle supplies a root between each pair of distinct roots. Hence

$$N(Q) - 2 \leq N(Q'') \leq 2(m - 1) - 1, \quad (\text{S47})$$

so $N(P_m) \leq 2m - 1$; the cases $N < 2$ are immediate. In Eq. (S45), every affine coefficient has nonzero slope $-w_j e^{-c_j^2/(2\sigma^2)}$, so the claim applies. $\square$

Lemma S3.4 (Uniform adjacent isolation). *For the fixed finite centre set and the compact weight family, there are $C_{\text{iso}}, q_{\text{iso}} > 0$ such that, for every $j < m$,*

$$\sup_{x \in [c_j, c_{j+1}]} \frac{\sum_{k \notin \{j, j+1\}} q_k(x)}{q_j(x) + q_{j+1}(x)} \leq C_{\text{iso}} e^{-q_{\text{iso}}/\sigma^2} \quad (\text{S48})$$

for all sufficiently small σ , uniformly in w . On $[\alpha, c_1]$, component 1 has the analogous one-component bound, and on $[c_m, \beta]$, component m does.

Proof. For $x \in [c_j, c_{j+1}]$ and $k < j$,

$$(x - c_k)^2 - (x - c_j)^2 = (c_j - c_k)(2x - c_j - c_k) \geq (c_j - c_k)^2. \quad (\text{S49})$$

For $k > j + 1$, comparison with c_{j+1} gives the corresponding positive lower bound. Since $w_k/w_j \leq 1/w_*$, summation over the finite centre set proves Eq. (S48). The same difference-of-squares comparison proves the two tail statements. The posterior mean and variance of the full mixture therefore differ from those of the isolated adjacent pair by $O(e^{-q/\sigma^2})$, uniformly in the gap, the allocation, and sufficiently small σ . $\square$

Theorem S3.5 (Exact topology of the pure mixture). *Uniformly for $w \in \mathcal{W}_{w_*}$, every sufficiently small $\sigma > 0$ gives exactly m simple local maxima and $m - 1$ simple local minima of $H_{\sigma, w}$ on J , ordered alternately, with nonzero endpoint derivatives. The maximum near c_j , denoted $p_j(\sigma, w)$, satisfies*

$$p_j = c_j + O(e^{-q/\sigma^2}), \quad L'_{\sigma, w}(p_j) = -\sigma^{-2} + o(\sigma^{-2}). \quad (\text{S50})$$

For $m \geq 2$, define the weighted equality point

$$s_j(\sigma, w) = \frac{c_j + c_{j+1}}{2} + \frac{\sigma^2}{\Delta_j} \log \frac{w_j}{w_{j+1}}. \quad (\text{S51})$$

The minimum in the j -th gap, denoted $r_j(\sigma, w)$, satisfies

$$r_j = s_j + O(\sigma^4), \quad (\text{S52})$$

$$L'_{\sigma, w}(r_j) = \frac{\Delta_j^2}{4\sigma^4} + O(\sigma^{-2}). \quad (\text{S53})$$

All remainders are uniform on the compact weight set.

Proof. Fix a constant $A > 1$. On $|x - c_j| \leq A\sigma^2$, one-component isolation gives

$$L_{\sigma, w}(x) = \frac{c_j - x}{\sigma^2} + O(\sigma^{-2} e^{-q/\sigma^2}), \quad (\text{S54})$$

$$L'_{\sigma, w}(x) = -\sigma^{-2} + O(\sigma^{-4} e^{-q/\sigma^2}). \quad (\text{S55})$$

At $c_j - \sigma^2$ the slope is positive, at $c_j + \sigma^2$ it is negative, and it is strictly decreasing between them. Hence there is one simple maximum with the location and derivative in Eq. (S50).

For a gap, the isolated adjacent odds are exactly

$$\frac{q_{j+1}(x)}{q_j(x)} = \exp \left[\frac{\Delta_j(x - s_j)}{\sigma^2} \right]. \quad (\text{S56})$$

Put $h_j = (\log 9)/\Delta_j$ and

$$C_j = [s_j - h_j\sigma^2, s_j + h_j\sigma^2]. \quad (\text{S57})$$

At the left and right endpoints the adjacent posterior masses are, respectively, $(9/10, 1/10)$ and $(1/10, 9/10)$. By Lemma S3.4, after reducing a common σ_0 , both adjacent posterior masses are bounded below throughout C_j , and

$$\text{Var}_{\pi(x)}(c) \geq \kappa_0 \Delta_*^2, \quad L'_{\sigma,w}(x) \geq \frac{\kappa_v}{\sigma^4} > 0 \quad (x \in C_j) \quad (\text{S58})$$

for constants independent of j, w, σ . At the two endpoints of C_j , Eq. (S43) gives, for sufficiently small σ ,

$$L_{\sigma,w}(s_j - h_j\sigma^2) \leq -\frac{\Delta_j}{5\sigma^2}, \quad L_{\sigma,w}(s_j + h_j\sigma^2) \geq \frac{\Delta_j}{5\sigma^2}. \quad (\text{S59})$$

Thus C_j contains exactly one simple minimum root.

At s_j , the adjacent posterior masses are equal and the adjacent posterior mean is $(c_j + c_{j+1})/2$. Hence

$$L_{\sigma,w}(s_j) = -\frac{1}{\Delta_j} \log \frac{w_j}{w_{j+1}} + O(\sigma^{-2} e^{-q/\sigma^2}) = O(1) \quad (\text{S60})$$

uniformly in w . The mean-value theorem and Eq. (S58) give $r_j - s_j = O(\sigma^4)$. At r_j , the two adjacent posterior masses are $1/2 + O(\sigma^2)$; substitution in Eq. (S44) proves Eq. (S53).

For small σ , the constructed peak and gap neighbourhoods are pairwise disjoint and contained in (α, β) . They contain $m + (m - 1) = 2m - 1$ distinct simple stationary points. The global bound in Lemma S3.3 excludes every additional real stationary point, including multiple ones. Tail isolation and the fixed separations $c_1 - \alpha > 0$ and $\beta - c_m > 0$ give

$$L_{\sigma,w}(\alpha) > 0, \quad L_{\sigma,w}(\beta) < 0, \quad (\text{S61})$$

so neither endpoint is stationary. When $m = 1$, there are no gaps or valley definitions: the exact identity $L_{\sigma,w}(x) = (c_1 - x)/\sigma^2$ gives one maximum and both endpoint signs directly. $\square$

S3.3. Full posterior-sector certificate and a slow positive factor

The zero bound above does not apply after multiplication by a nonconstant positive factor. We therefore certify the complete complement of small peak and valley boxes.

Lemma S3.6 (Posterior-sector certificate). *Fix $K > 0$. There are constants $A_p, A_v, C_p, C_v, \kappa_p, \kappa_v > 0$ and $\sigma_K > 0$, depending only on the fixed centre set, J, w_* , and K , such that, for every $0 < \sigma < \sigma_K$ and every admissible w , the boxes*

$$P_j = [c_j - A_p\sigma^2, c_j + A_p\sigma^2], \quad (\text{S62})$$

$$V_j = [r_j - A_v\sigma^4, r_j + A_v\sigma^4] \quad (j < m) \quad (\text{S63})$$

are pairwise disjoint and contained in (α, β) , and the following hold:

$$|L_{\sigma,w}| \leq C_p, \quad L'_{\sigma,w} \leq -\kappa_p\sigma^{-2} \quad \text{on } P_j, \quad (\text{S64})$$

$$|L_{\sigma,w}| \leq C_v, \quad L'_{\sigma,w} \geq \kappa_v\sigma^{-4} \quad \text{on } V_j. \quad (\text{S65})$$

At the left and right box boundaries, respectively, the slope signs are

	left boundary	right boundary
P_j	$L_{\sigma,w} \geq 2K$	$L_{\sigma,w} \leq -2K$
V_j	$L_{\sigma,w} \leq -2K$	$L_{\sigma,w} \geq 2K$

(S66)

On the entire complement in J of the open boxes,

$$|L_{\sigma,w}(x)| \geq 2K, \quad (\text{S67})$$

with the exhaustive alternating sign order

$$+ P_1 - V_1 + P_2 - \cdots + P_m - . \quad (\text{S68})$$

For $m = 1$, every valley clause is omitted.

Proof. Choose $A_p \geq 4K$. The uniform expansions (S54)–(S55) give, after reducing σ_K , Eq. (S64) and the peak row of Eq. (S66). The convex-hull property $\bar{c}(x) \in [c_1, c_m]$ gives the full outer-tail estimates without any local expansion:

$$\begin{aligned} x \leq c_1 - A_p \sigma^2 &\implies L_{\sigma,w}(x) \geq A_p, \\ x \geq c_m + A_p \sigma^2 &\implies L_{\sigma,w}(x) \leq -A_p. \end{aligned} \quad (\text{S69})$$

Fix a gap and abbreviate $\Delta = \Delta_j$, $s = s_j$. The right adjacent posterior mass is

$$p(x) = \frac{q_{j+1}(x)}{q_j(x) + q_{j+1}(x)} = \frac{1}{1 + \exp[-\Delta(x - s)/\sigma^2]}. \quad (\text{S70})$$

On the crossover interval C_j in Eq. (S57), its edge odds are $1/9$ and 9 , not exponentially small or large. Both adjacent masses are therefore bounded below throughout C_j , and Eq. (S58) holds there. Conversely, the fixed finite centre diameter bounds the posterior variance, so $|L'| \leq C\sigma^{-4}$ on C_j .

The pure root satisfies $L(r_j) = 0$. Choose A_v large enough that $\kappa_v A_v \geq 2K$. Since $A_v \sigma^4 = o(\sigma^2)$, the valley box lies in C_j for small σ . Integrating the lower and upper bounds for L' from r_j proves Eq. (S65), the valley row of Eq. (S66), and the required negative and positive bounds on $C_j \setminus \text{int } V_j$.

It remains to cover both outer sectors of every gap; this is the step that prevents an unproved dominance claim at a crossover edge. Consider first

$$[c_j + A_p \sigma^2, s_j - h_j \sigma^2] \quad (\text{S71})$$

and split it at $c_j + \Delta_j/4$, which lies in the stated interval for all sufficiently small σ , uniformly in w . On the portion nearer c_j , every other component is exponentially small relative to component j , so

$$\bar{c}(x) - x \leq -\frac{1}{2}A_p \sigma^2, \quad L_{\sigma,w}(x) \leq -A_p/2 \leq -2K. \quad (\text{S72})$$

On the remaining portion, monotonicity of Eq. (S70) and the left crossover edge give $p(x) \leq 1/10$, while $x - c_j \geq \Delta_j/4$. The adjacent-pair mean then obeys

$$c_j + \Delta_j p(x) - x \leq \frac{\Delta_j}{10} - \frac{\Delta_j}{4} = -\frac{3\Delta_j}{20}. \quad (\text{S73})$$

The nonadjacent correction is exponentially small by Lemma S3.4; hence this sector also has $L \leq -2K$ for small σ , including its $1/9$ -odds edge.

Symmetrically, split $[s_j + h_j \sigma^2, c_{j+1} - A_p \sigma^2]$ at $c_{j+1} - \Delta_j/4$. On the portion nearer the crossover, $p(x) \geq 9/10$ and $c_{j+1} - x \geq \Delta_j/4$, giving the positive analogue of Eq. (S73); on the portion nearer c_{j+1} , one-component isolation gives $L \geq A_p/2 \geq 2K$. Thus the right outer sector has $L \geq 2K$.

The fixed centre and endpoint separations make all boxes disjoint and interior for one common sufficiently small σ_K . The outer tails, all peak boundaries, both outer sectors of every gap, and every crossover-minus-valley sector cover every point of the box complement. This proves Eqs. (S67)–(S68). When $m = 1$, the same result follows directly from $L = (c_1 - x)/\sigma^2$ and Eq. (S69). $\square$

Theorem S3.7 (Slow positive factors preserve the exact topology). *Let $x \in C^2(I)$ be strictly increasing with $\inf_I x' = v_0 > 0$. Let $a_\sigma \in C^2(I)$ be positive and assume*

$$M_0 := \sup_{\sigma < \sigma_0} \|\partial_t \log a_\sigma\|_{L^\infty(I)} < \infty, \quad M_1 := \sup_{\sigma < \sigma_0} \|\partial_t^2 \log a_\sigma\|_{L^\infty(I)} < \infty. \quad (\text{S74})$$

Then, uniformly for $w \in \mathcal{W}_{w_}$, all sufficiently small $\sigma > 0$ give exactly m nondegenerate maxima and $m-1$ nondegenerate minima of*

$$F_{\sigma,w}(t) = a_\sigma(t)H_{\sigma,w}(x(t)) \quad (\text{S75})$$

on I , ordered alternately, with no stationary endpoint. Relative to the pure roots in the x coordinate,

$$x(t_{j,\sigma}^{\max}) - p_j(\sigma, w) = O(\sigma^2), \quad (\text{S76})$$

$$x(t_{j,\sigma}^{\min}) - r_j(\sigma, w) = O(\sigma^4), \quad (\text{S77})$$

uniformly in w . Thus

$$x(t_{j,\sigma}^{\min}) = s_j(\sigma, w) + O(\sigma^4). \quad (\text{S78})$$

Proof. Put

$$b_\sigma = \partial_t \log a_\sigma, \quad D_{\sigma,w}(t) = \partial_t \log F_{\sigma,w}(t) = b_\sigma(t) + x'(t)L_{\sigma,w}(x(t)). \quad (\text{S79})$$

Since $F_{\sigma,w} > 0$, its stationary points are precisely the zeros of $D_{\sigma,w}$, and

$$D'_{\sigma,w} = b'_\sigma + x''L_{\sigma,w} + (x')^2 L'_{\sigma,w}. \quad (\text{S80})$$

Choose $K > (M_0 + 1)/v_0$ and apply Lemma S3.6. On the full complement of the open boxes, the term $x'L$ strictly dominates b_σ , so D has the alternating signs in Eq. (S68) and has no complement zero.

On a peak box, Eqs. (S64) and (S80) give

$$D'_{\sigma,w} \leq M_1 + \|x''\|_\infty C_p - v_0^2 \kappa_p \sigma^{-2} < 0 \quad (\text{S81})$$

for small σ . Its boundary signs give exactly one zero. At that zero $F'' = FD' < 0$, hence the root is a nondegenerate maximum. Similarly, on a valley box,

$$D'_{\sigma,w} \geq -M_1 - \|x''\|_\infty C_v + v_0^2 \kappa_v \sigma^{-4} > 0, \quad (\text{S82})$$

so its unique zero is a nondegenerate minimum. The certified tail signs exclude endpoint stationary points.

At a slow-factor root, Eq. (S79) gives $|L| \leq M_0/v_0$. The pure peak root has $L(p_j) = 0$, and the uniform negative $L' = \Theta(\sigma^{-2})$ on its box implies the peak displacement in Eq. (S76). The uniform positive $L' = \Theta(\sigma^{-4})$ on a valley box gives Eq. (S77). Combining the latter with Eq. (S52) proves Eq. (S78). Every constant is uniform over the compact allocation family. $\square$

S3.4. Fixed- ε Doi transfer and theorem boundary

For $B > 0$, let

$$f_{B,\varepsilon,w}(t) = \mathbb{E}_0 \left[K_{B,w,\varepsilon}(Z_t, R_t) \exp \left(- \int_0^t K_{B,w,\varepsilon}(Z_s, R_s) ds \right) \right] \quad (\text{S83})$$

be the Doi reaction-time density and set

$$\mathcal{F}_{B,\varepsilon,w} = f_{B,\varepsilon,w}/B. \quad (\text{S84})$$

This is budget-rescaling, not normalization to unit time integral.

Theorem S3.8 (Exact m Doi modes for fixed finite (d, m)). *Fix all data in Eqs. (S15)–(S35), including the stationary midpoint variance, mutual independence, the strict weighted-space variance inequalities, target times strictly inside I , consistently oriented physical centres, normalized slabs of physical width $\varepsilon\rho$, the whole-window contact margin, and the compact allocation family. Then there is $\varepsilon_0 > 0$, depending on these fixed data, such that for every $0 < \varepsilon < \varepsilon_0$ and every $w \in \mathcal{W}_{w*}$, the continuum free-exposure clock $G_{\varepsilon,w}$ has exactly m nondegenerate maxima and $m - 1$ nondegenerate minima on I , ordered alternately, and no stationary endpoint.*

For each fixed such ε , there is $B_0(\varepsilon) > 0$, uniform over $w \in \mathcal{W}_{w}$, such that $\mathcal{F}_{B,\varepsilon,w}$ has the same complete stationary signature for every $0 < B < B_0(\varepsilon)$. Multiplication by $B > 0$ changes no stationary point, so the statement also holds for the physical density $f_{B,\varepsilon,w}$.*

Proof. In Lemma S3.2, take $a_\sigma = a_\varepsilon$, with $\sigma = \varepsilon S_*/\ell_0$. Lemma S3.1 supplies Eq. (S74); hence Theorem S3.7 proves the exact stationary signature of $G_{\varepsilon,w}$, uniformly over the compact allocation family.

Fix one $0 < \varepsilon < \varepsilon_0$. The ordered simple roots depend continuously on w by the implicit-function theorem. Their finitely many graphs over the compact set $\mathcal{W}_{w_*}$ are compact and disjoint. Continuity of the second time derivative gives pairwise disjoint root tubes on which G'' has a uniform nonzero sign and magnitude. After shrinking the tubes if necessary, G' has fixed opposite signs at the two boundaries of each tube. On the compact complement in $I \times \mathcal{W}_{w_*}$, $|G'|$ has a positive minimum; the two endpoint slopes have positive uniform absolute minima as well.

For this fixed ε , $V_{w,\varepsilon}$ is bounded uniformly in w , and Eqs. (S18)–(S20) put q_0 in the required weighted state space. Because $w_j \geq w_* > 0$, the compact allocation set lies strictly inside the simplex and admits the affine control chart and a complex tube required by Theorem S2.1. (If the set is the singleton $w_j = 1/m$, the same conclusion is the corresponding pointwise case.) Applying that theorem with time orders 0, 1, 2 and no control derivative gives

$$\sup_{w \in \mathcal{W}_{w_*}} \|\mathcal{F}_{B,\varepsilon,w} - G_{\varepsilon,w}\|_{C^2(I)} \longrightarrow 0 \quad (B \downarrow 0). \quad (\text{S85})$$

Choose $B_0(\varepsilon)$ so that the C^1 error is smaller than all boundary, complement, and endpoint margins and the second-derivative error is smaller than the root-tube curvature margin. In each tube, $\mathcal{F}'$ has opposite boundary signs and $\mathcal{F}''$ keeps the sign of G'' , so it has exactly one root of the same type. The complement and endpoints contain none. This proves the complete finite-window signature. $\square$

Remark S3.9 (Quantifiers, saturation, and scope). *The quantifier order is*

$$\text{fix } (d, m, \text{all geometric and probabilistic data}) \implies 0 < \varepsilon < \varepsilon_0 \implies 0 < B < B_0(\varepsilon). \quad (\text{S86})$$

There is no asserted lower bound for $B_0(\varepsilon)$ uniform as $\varepsilon \downarrow 0$, and no interchange of the two limits. The construction is not uniform in d or m , does not provide one geometry with arbitrarily many modes, does not count stationary points outside I , and gives no process-level event-mass floor. It treats normalized longitudinal Gaussian slabs, not arbitrary localized catalyst patches.

Moreover, Lemma S3.1 shows that the relative contact factor is asymptotically saturated on the theorem window:

$$c_{d,\varepsilon} = 1 + O(\varepsilon^{-N} e^{-q/\varepsilon^2}) \quad \text{in } C^2(I). \quad (\text{S87})$$

Thus the exact Doi embedding is rigorous, but the modal basis is created by the ordered narrow slabs sampled by the monotone constant-variance midpoint; the relative encounter factor is a slow perturbation in this asymptotic family. Nontrivial contact, a useful common positive budget, deterministic finite-parameter complement certificates, survival, and event-basin masses remain separate continuum-validation requirements rather than consequences of Theorem S3.8.

S4. SCOPE OF THE ANALYTICAL RESULT

The proof establishes an exact existence theorem for the continuum Doi process. It does not use a finite-volume approximation, and no discretized convergence statement is needed for its conclusion. A separate claim that numerically located stationary points converge from bounded grids to the unbounded continuum would require quantitative spatial and truncation errors together with componentwise preservation of the root margins. No such numerical claim is made here.

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